Chapter 2: 1.35, 2.1, 5, 10, 16, 20, 22, 31, 41, 43, 44
2.7 + 2.8

for a sphere in air at STP

2.1

$$\frac{f_{quad}}{f_{lin}} = \frac{cv^2}{bv} = \frac{\gamma D}{\beta} v = \left(1.6 \cdot 10^3 \frac{s}{m^2}\right) \cdot D \cdot v$$

basketball → $D = 7\,cm$

$$\frac{f_{quad}}{f_{lin}} = 1 \leftarrow \text{so find velocity}$$

$$1 = 1.6 \cdot 10^3 \frac{s}{m^2} \cdot (0.07\,m) \cdot v$$

$$v = 0.0089\,m/s \rightarrow \approx 1\,cm/s$$

beachball → $D = 70\,cm$

$$\hookrightarrow v = \frac{1}{1.6 \cdot 10^3 \cdot (0.7\,m)} = 0.00089\,m/s \approx 1\,mm/s$$

both are dominated by quadratic drag.

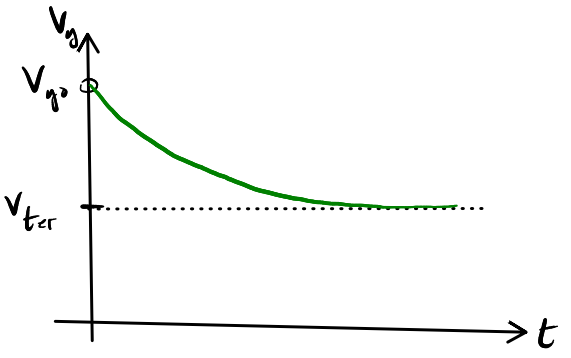
2.5 $v_{y0} > v_{ter}$

$$mg - bv = m\frac{dv}{dt}$$

need to solve this ODE →

$$\frac{dv}{dt} + \frac{b}{m}v = g \Rightarrow v_{term} = \frac{mg}{b}$$

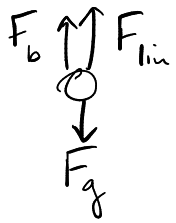
$$v(t) = \frac{mg}{b} + Ce^{-bt/m}$$



2.7 need to solve w/ separation of variables just like in problem 2.8 + 2.9

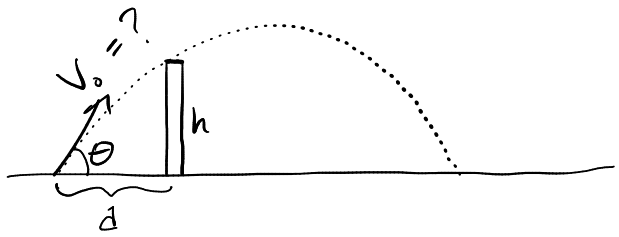
2.8 just integrate and algebra

2.10



$$\Sigma F = -F_b - F_{lin} + F_g = m_{st} a$$

2.16



$$\Sigma F_y = ma_y \qquad \rightarrow V_{yi} = V_0 \sin\theta$$

$$-mg = ma_y \qquad \rightarrow V_x = V_0 \cos\theta$$

solve this DE → $-g = a_y = \ddot{y} = \frac{d^2y}{dt^2}$

w/ $y_i = 0$ → $y(t) = V_{iy}t + \frac{1}{2}a_y t^2$

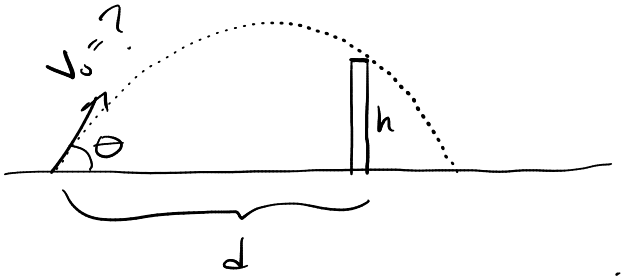
$$h = V_0 \sin\theta \cdot \frac{d}{V_0 \cos\theta} - \frac{1}{2}g\left(\frac{d}{V_0\cos\theta}\right)^2$$

$$h = \tan\theta\, d - \frac{g}{2}\frac{d^2}{V_0^2\cos^2\theta}$$

$$\frac{g}{2}\frac{d^2}{V_0^2\cos^2\theta} = (\tan\theta\, d - h)$$

$$\frac{gd^2}{2\cos^2\theta} = V_0^2(\tan\theta\, d - h)$$

$$\boxed{V_0 = \sqrt{\frac{gd^2}{2\cos^2\theta(d\tan\theta - h)}}}$$



$$x(t) = v_x \cdot t$$

$$x(t) = d = v_x \cdot t$$

$$t = \frac{d}{v_x} = \frac{d}{V_0\cos\theta}$$

$d = 50\text{ m}$

$h = 2$

$\theta = 25^\circ$

$$V_0 = \sqrt{\frac{9.8\cdot(50)^2}{2\cos^2(25)(50\tan 25 - 2)}}$$

$$V_0 = 26.45\text{ m/s}$$

if $\tan\theta < \frac{h}{d}$

then our solution will be in trouble

$$V_0 = \sqrt{\frac{gd^2}{d\cdot 2\cos^2\theta\left(\frac{d}{d}\tan\theta - \frac{h}{d}\right)}} = \sqrt{\frac{gd}{2\cos^2\theta\left(\tan\theta - \frac{h}{d}\right)}}$$

if $\tan\theta < \frac{h}{d}$ then V_0 is imaginary

solve for d instead and find the two distances that the ball will clear

2.20

$$x(t) = v_{x0}\,\tau \cdot \left(1 - e^{-t/\tau}\right)$$

$$y(t) = (v_{y0} + v_t)\,\tau \cdot \left(1 - e^{-t/\tau}\right) - v_{ter}\cdot t$$

linear drag → $m\ddot{\vec{r}} = m\vec{g} - bv$

$v_{x0} = v_0 \cdot \cos\theta \leftarrow 45°$

$v_{y0} = v_0 \cdot \sin\theta$

$$\boxed{y = \frac{v_{y0} + v_t}{v_{x0}} \cdot x + v_{ter}\cdot\tau \ln\left(1 - \frac{x}{v_{x0}\cdot\tau}\right)}$$

$\tau = \frac{m}{b}$

$v_{ter} = \frac{mg}{b}$

$\boxed{v_{ter} = \tau \cdot g}$

$\tau = \frac{v_{ter}}{g}$

plot $v_{ter} = 0.3, 1, 3, \infty$

$$y = \frac{v_{y0} + v_t}{v_{x0}} \cdot x + \frac{v_{ter}^2}{g} \cdot \ln\left(1 - \frac{x\cdot g}{v_{x0}\cdot v_{ter}}\right)$$

- $x = v_{x0}\cdot t \Rightarrow t = \boxed{\frac{x}{v_{x0}}}$
- $y = v_{y0}\cdot t + \frac{1}{2}a_y t^2$

$$\boxed{y = \frac{v_{y0}}{v_{x0}}\cdot x + \frac{1}{2}a_y\left(\frac{x}{v_{x0}}\right)^2}$$

```
x = np.linspace(0, 100, 1000)
                ↑
def y(x, v_x0, v_y0, v_t):
    return(                    )
```

```
def y(x, v_x0, v_y0):
```

```
ax.plot(x, y(x, v_x0, v_y0, 0.3))
ax.plot(x, y(x, v_x0, v_y0, 1))
ax.plot(x, y(x, v_x0, v_y0, 3))
```

2.31 a) $V_{ter} = \sqrt{\frac{Mg}{C}}$ $\quad C = \gamma D^2$ $\quad \gamma_{air} = 0.25 \frac{Ns^2}{m^4}$

ax.plot(x, y(x, v_{x0}, v_{y0}, 10°))

$$C = 0.25 \cdot (0.24\,m)^2$$

$$V_{ter} = \sqrt{\frac{0.6kg \cdot 9.8 N/kg}{0.25 (0.24)^2}} = \boxed{20.2\,m/s}$$

b) $h = 30\,m$